Ankitha Betala

Structural Analysis Engineer
Aerospace Engineer

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Academic Qualifications

Year	Degree/Certificate	Institute	CPI/%
2023 - 2026	M.Tech (Aero Thermal)	Indian Institute of Technology, Kanpur	6.5/10
2023	B. E (Aeronautical Engg)	Hindusthan Institute of Technology, coimbatore	8.86/10
2019	Board of Intermediate Education(XII)	Sri chaitanya Jr.Mahila Kalasala,Vijayawada	9.92/10
2017	Board of Secondary Education(X)	Andhra pradesh Residential School,Musunuru	9.5/10

Technical Skills

- **FEA:** ANSYS Mechanical, Static Structural Analysis, Thermal Analysis, Fluid-Structure Interaction (FSI)
- **HPC & Computing:** High Performance Computing (HPC), Linux Environment, Remote Cluster Computing, MobaXterm
- **Tools:** ANSYS Workbench, NI LabVIEW

Professional Experience

Engineer II (*Jan'26- Present*)
Offered by: INVENTGRID INDIA PRIVATE LIMITED (IG Defence)
Office: First Floor, D-23, Sector 63, Noida, Delhi-NCR 201309 Department: New Development – CFD Analysis

JWALA Defence Project

- Performed linear and nonlinear static structural analyses of rocket motor components using ANSYS Mechanical to evaluate stress distribution and deformation under internal pressure and applied loads.
- Defined material properties, boundary conditions, contact settings and load cases to assess structural response under operating conditions.
- Evaluated von Mises stress, total deformation and critical stress concentrations to identify potential structural failure locations.
- Analysed threaded connections and contact interfaces, investigating relative sliding, contact separation and potential component detachment.
- Conducted fluid–structure interaction (FSI) analysis of a rocket nozzle to evaluate the effects of internal pressure loading on structural deformation and stress.
- Prepared technical reports and engineering documentation.

KAL RATO Booster Project

- Post-processed KAL RATO test data to extract key performance parameters and assess booster performance for a 250 kg payload application.
- Calculated required thrust and total impulse for rail inclinations of 15°–20° and target release speeds of 160–190 km/h.
- Performed linear and nonlinear static structural analyses of rocket motor components using ANSYS Mechanical to evaluate stresses and deformation.
- Analysed threaded connections and contact interfaces to assess potential sliding, separation, and component detachment.
- Conducted structural analysis of the motor test rig to evaluate stress distribution and potential failure locations under test loads.
- Performed FSI and thermal analyses to assess pressure-induced deformation and temperature distribution in motor components.

Research Interests

- Structural Analysis, Finite Element Methods (FEM), and Nonlinear Contact Mechanics
- Thermal–Structural Analysis and Thermomechanical Behaviour
- Fluid–Structure Interaction (FSI) and Pressure-Induced Deformation
- Rocket Motor and Nozzle Structural Integrity
- Stress Analysis, Failure Assessment, and Design Optimisation
- Gas Turbine Components, Thermal Management, and Heat Transfer

Key Projects

M.Tech Thesis: Conjugate Heat Transfer Analysis of a Circular Nozzle with Passive Cooling (*Aug'2024 - 2025*)
Proposed by: Aeronautical Development Establishment (ADE), DRDO
Supervisor: Dr. Abhijith Kushari — Indian Institute of Technology Kanpur

- Conducted numerical investigation of conjugate heat transfer in a circular nozzle with passive cooling using **ANSYS Fluent**.
- Studied the coupled effects of reacting flow, heat conduction, and convection to evaluate nozzle thermal performance.

- Analyzed temperature distribution, heat flux, cooling effectiveness, and thermal gradients within the nozzle structure.
- Performed simulations on High Performance Computing (HPC) clusters using Linux-based remote computing environments.

Welding of Dissimilar Aluminium Alloys (Al7075-Al6063) (*Jan'23 – May'23*)
Queen India Engineering Services, Chennai

- Investigated TIG welding of dissimilar aluminium alloys using ER4043 filler material.
- Evaluated mechanical properties through hardness, impact, and microstructural analysis.
- Performed non-destructive testing (NDT) to identify surface and internal defects.

Course Projects

- **Temperature Measurement of Wind Driven pool fire using Labview** (*April 15,2024*)
  Guide: Dr.Kamal Poddar | Department of Aerospace Engineering, Indian Institute of Technology,Kanpur.
  - Developed a LabVIEW interface for temperature acquisition and monitoring in pool fire experiments.
  - Performed data acquisition, processing, and temperature measurement analysis.
- **Transient Thermal Analysis Of a Sphere With an Enclosure** (*Jan'2024-May'2024*)
  Guide:,DR. MOHAMMAD IBRAHIM SUGARNO | Department of Aerospace Engineering, IIT Kanpur.
  - Conducted transient thermal analysis using ANSYS Workbench.
  - Investigated temperature evolution and heat transfer characteristics of a solid sphere enclosed by an air domain.
- **Finite Element Analysis Convergence and Mesh Independence** | Certificate

Internships

Solid Rocket Motor Thrust Vector Control (*Oct'22- 2023*)
Offered by: Defence Research and Development Organisation(DRDO)
Guide: SV Raja Goutham,Scientist 'E',Advanced Systems Lab, DRDO ,Ministry of Defence, Govt.of India ,Hyderabad-500058

- Acquired Theoritical and Analytical Knowledge in Rocket Propulsion
- Understand solid rocket motor dynamics and thrust vectoring principles.
- Enhance rocket trajectory control and stability.
- During the internship programme,I gained knowledge about how pressure varies at different points of the nozzle.
- designed elliptical convergent divergent nozzle in autocad and subdivided into three parts.
- studied the methods of thrusting vectoring and how they are operated

Positions of Responsibility

- **Teaching Assistant,Engineering Graphics (Indian Institute of Technology, Kanpur)** (*Aug'2024-Jul2025*)
  - Worked as a Teaching Assistant in Engineering Graphics, compulsory for the junior undergraduates.
  - Responsible for the design of written and computer-based assignments for the course.
  - Mentored 30 UG students and reviewed their continuous progress
- **Volunteered in ACGT Conference** (*Aug'2024*)
  - Guided **80** participants in acclimatizing to the Environment of the Institute.

Scholastic Achievements

- Secured **All India Rank 1017** in **Gate 2023.**
- Received the **Prathibha puraskar award in 2017 on SSC merit, by Government of Andhra pradesh.** in 2017.
- Received the **Ugam legrand scholarship on Intermediate merit basis,2019-2023.**